

Higher Order Linear ODEs

1 General Theory

Theorem 1.1 (Existence and Uniqueness Theorem).

Let $I \subseteq \mathbb{R}$ be an open interval and suppose functions $g, p_0, p_1, \dots, p_{n-1}$ are continuous on I . Then for $t_0 \in I$ and real numbers $y_0, y_1, \dots, y_{n-1}$, there exists exactly one solution to the IVP

$$\begin{aligned} y^{(n)} + p_{n-1}(t)y^{(n-1)} + \dots + p_1(t)y' + p_0(t)y &= g(t), \\ y(t_0) = y_0, \quad y'(t_0) = y_1, \quad \dots, \quad y^{(n-1)}(t_0) &= y_{n-1}. \end{aligned}$$

Theorem 1.2 (Principle of Superposition).

If $y_1, y_2, \dots, y_n$ are solutions to the homogeneous linear ODE

$$y^{(n)} + p_{n-1}(t)y^{(n-1)} + \dots + p_1(t)y' + p_0(t)y = 0,$$

then any linear combination

$$y = c_1y_1 + c_2y_2 + \dots + c_ny_n,$$

where $c_1, c_2, \dots, c_n$ are constants, is also a solution to the homogeneous equation.

Definition 1.1 (Wronskian).

The Wronskian of n functions $y_1, y_2, \dots, y_n$ is defined as the determinant

$$W(y_1, y_2, \dots, y_n)(t) = \begin{vmatrix} y_1(t) & y_2(t) & \dots & y_n(t) \\ y_1'(t) & y_2'(t) & \dots & y_n'(t) \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)}(t) & y_2^{(n-1)}(t) & \dots & y_n^{(n-1)}(t) \end{vmatrix}.$$

Definition 1.2 (Fundamental Set of Solutions).

The set of n solutions $y_1, y_2, \dots, y_n$ to an n -th order homogeneous linear ODE defined on an interval I is called a **fundamental set of solutions (FSS)** if every solution to the homogeneous equation can be expressed as a linear combination of $y_1, y_2, \dots, y_n$.

Theorem 1.3.

If $p_0, \dots, p_{n-1}$ are continuous on an interval I and $y_1, \dots, y_n$ are solutions to the homogeneous linear ODE

$$y^{(n)} + p_{n-1}(t)y^{(n-1)} + \dots + p_1(t)y' + p_0(t)y = 0,$$

then every solution to the ODE can be expressed as a linear combination of $y_1, \dots, y_n$ if and only if

$$\exists t \in I \text{ s.t. } W(y_1, \dots, y_n)(t) \neq 0$$

Theorem 1.4.

Let $y_1, \dots, y_n$ be the solutions to the homogeneous linear ODE

$$y^{(n)} + p_{n-1}(t)y^{(n-1)} + \dots + p_1(t)y' + p_0(t)y = 0, \quad \text{for } t \in I.$$

Then

$$y_1, \dots, y_n \text{ are linearly independent} \iff \exists t \in I \text{ s.t. } W(y_1, \dots, y_n)(t) \neq 0.$$

Theorem 1.5 (Abel's Theorem).

If $y_1, \dots, y_n$ are solutions to the homogeneous linear ODE

$$y^{(n)} + p_{n-1}(t)y^{(n-1)} + p_{n-2}(t)y^{(n-2)} + \dots + p_1(t)y' + p_0(t)y = 0, \quad \text{for } t \in I,$$

then the Wronskian is given by

$$W(y_1, \dots, y_n)(t) = c \cdot \exp\left(-\int p_{n-1}(t) dt\right) \quad \text{where } c \text{ is a constant depending on } y_1, \dots, y_n.$$

2 Homogeneous Equations with Constant Coefficients

For constants $a_n \neq 0, a_{n-1}, \dots, a_1, a_0 \in \mathbb{R}$,

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = 0$$

Characteristic polynomial:

$$\begin{aligned} Z(r) &= a_n r^n + a_{n-1} r^{n-1} + \dots + a_1 r + a_0 \\ &= a_n (r - r_1)(r - r_2) \dots (r - r_n) \\ &= a_n (r - r_1)^{m_1} (r - r_2)^{m_2} \dots (r - r_k)^{m_k} \end{aligned}$$

where $r_1, r_2, \dots, r_n \in \mathbb{C}$ are the roots of $Z(r)$.

Complex-valued solutions

Complex roots must appear in conjugate pairs, i.e., if $r = \lambda + i\mu$ is a root, then its conjugate $\bar{r} = \lambda - i\mu$ is also a root. The complex-valued solution pair $y = e^{(\lambda+i\mu)t}$ and $y = e^{(\lambda-i\mu)t}$ can be re-expressed as the real-valued solution pair:

$$y = e^{\lambda t} \cos(\mu t) \quad \text{and} \quad y = e^{\lambda t} \sin(\mu t).$$

3 Non-Homogeneous Equations

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = g(t)$$

Then the general solution is given by

$$y(t) = y_c(t) + y_p(t),$$

where y_c is the complementary solution (homogeneous solution) and y_p is a particular solution.

3.1 Method of Undetermined Coefficients

3.2 Variation of Parameters

Theorem 3.1 (Variation of Parameters).

Suppose $g, p_0, \dots, p_{n-1}$ are continuous on an interval I . Then a particular solution to the non-homogeneous linear ODE

$$y^{(n)} + p_{n-1}(t)y^{(n-1)} + \dots + p_1(t)y' + p_0(t)y = g(t)$$

can be found by

$$y_p(t) = u_1(t)y_1(t) + u_2(t)y_2(t) + \dots + u_n(t)y_n(t)$$

where $y_1, y_2, \dots, y_n$ is a FSS to the corresponding homogeneous equation and $u_1, u_2, \dots, u_n$ are functions given by

$$u_i(t) = \int \frac{g(t) \cdot \det[\mathbf{M}_i(t)]}{W(y_1, \dots, y_n)(t)} dt, \quad \text{for } i = 1, 2, \dots, n, \quad \text{and}$$

$$\mathbf{M}_i = \begin{pmatrix} y_1 & \dots & y_{i-1} & 0 & y_{i+1} & \dots & y_n \\ y_1' & \dots & y_{i-1}' & 0 & y_{i+1}' & \dots & y_n' \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)} & \dots & y_{i-1}^{(n-1)} & 1 & y_{i+1}^{(n-1)} & \dots & y_n^{(n-1)} \end{pmatrix}.$$

Let $u_1, u_2, \dots, u_n$ be functions satisfying the system of equations

$$\begin{pmatrix} y_1 & y_2 & \cdots & y_n \\ y'_1 & y'_2 & \cdots & y'_n \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-2)} & y_2^{(n-2)} & \cdots & y_n^{(n-2)} \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_n^{(n-1)} \end{pmatrix} \begin{pmatrix} u'_1 \\ u'_2 \\ \vdots \\ u'_{n-1} \\ u'_n \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ g(t) \end{pmatrix}$$

Then it can be verified that $y_p = u_1 y_1 + \cdots + u_n y_n$ is a particular solution to the non-homogeneous equation.